

Smart Buy Advisor v2.0: Full Cash vs. EMI + Salary Pledge & Real Effective Cost Calculator

Subject Code: 23EC5512 — Capstone Project

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1. ABSTRACT

In modern retail finance, consumers frequently evaluate purchasing decisions purely on immediate cash outlay rather than considering the opportunity cost of capital.

Smart Buy Advisor v2.0 extends conventional financial modeling by introducing a **Salary EMI Pledge Engine (0% to 100%)** alongside initial capital investment compound simulations.

Built using React 18, JavaScript ES6, and Web Storage API, this application simulates how pledging monthly EMI payments directly from salary keeps initial savings capital 100% intact, maximizing compound wealth accumulation over the tenure.

The application is compiled for static client-side execution and hosted live on GitHub Pages.

2. PROBLEM STATEMENT

Consumers possessing sufficient cash to buy high-value items upfront often pay 100% cash to avoid perceived debt. In doing so, they sacrifice compound investment returns on their savings capital.

Conversely, those opting for EMIs often fail to calculate the compounding advantage of leaving principal capital untouched when monthly EMIs are partially or fully serviced via regular salary income.

There is a lack of intuitive, real-time comparison tools visualizing this salary-backed compounding acceleration and the resulting real effective out-of-pocket cost of the purchase.

3. PROJECT OBJECTIVES

- Interactive Inputs:** Configure purchase price, down payment, EMI tenure, EMI scheme type (Standard vs. No-Cost), investment ROI, Salary Pledge %, and Extra Capital Injections.
- v2.0 Financial Engine:** Develop a modular JS financial engine simulating compound fund balances under variable salary contributions (0% to 100%).
- Real Effective Cost Architecture:** Calculate the exact real out-of-pocket cost of the item by deducting investment interest returns earned from nominal purchase outflows.
- Static GitHub Pages Hosting:** Compile the application into static HTML/JS/CSS assets for offline and online hosting on GitHub Pages with zero server setup.

4. TECHNICAL ARCHITECTURE & TECH STACK

- **User Interface Layer:** Built using React 18 functional components, custom hooks (useState, useMemo, useEffect), range sliders, and CSS Grid/Flexbox layouts.
- **Financial Engine:** Pure JavaScript module (calculatorEngine.js) implementing standard banking EMI formulas and monthly compound schedule iterations.
- **Web Storage Layer:** LocalStorage wrapper service (storageService.js) facilitating client-side CRUD scenario management.
- **Build Pipeline:** Compiled with Vite bundler (base: './') exporting optimized production static assets.

5. MATHEMATICAL METHODOLOGY & FORMULAS

- **Standard Banking EMI Formula:**

$$E = [P \times r \times (1+r)^n] / [(1+r)^n - 1]$$

- **Salary-Pledged Investment Schedule:**

$$B_t = B_{t-1} \times (1 + m) - [E \times (1 - S/100)]$$

- **Real Effective Out-of-Pocket Cost:**

$$Real\ Effective\ Cost = Original\ Outflow\ (Down\ Payment + Fees + EMIs) - Investment\ Returns\ Earned\ (R_{earned})$$

- **Net Money Saved & Effective Discount %:**

$$Net\ Savings = Full\ Cash\ Price - Real\ Effective\ Cost$$

$$Effective\ Discount\ \% = (Net\ Savings / Full\ Cash\ Price) \times 100$$

6. CASE STUDY & RESULTS ANALYSIS

Financial Metric	Option A (Full Cash)	Option B (100% Salary Pledge)	Net Advantage
Original Purchase Outflow	■1,10,000	■1,10,000	Same Cash Outflow
Investment Returns Earned	■0	-■43,077 (Offset)	■43,077 Interest Subvention
Real Effective Cost Out-of-Pocket	■1,10,000	■66,923	■43,077 Less Out of Pocket
Effective Discount Realized	0% OFF	39.2% OFF	■43,077 Net Discount!

7. CONCLUSION & LIVE LINKS

Smart Buy Advisor v2.0 successfully transforms retail purchase decision-making by demonstrating how opportunity returns subsidize the real out-of-pocket cost of high-value items.

■ **Live Application URL:** <https://nitintimothy.github.io/capstone-project-23EC5512/smart-buy-advisor/>

■ **GitHub Repository:** [git@github.com:NitinTimothy/capstone-project-23EC5512.git](https://github.com/NitinTimothy/capstone-project-23EC5512.git)